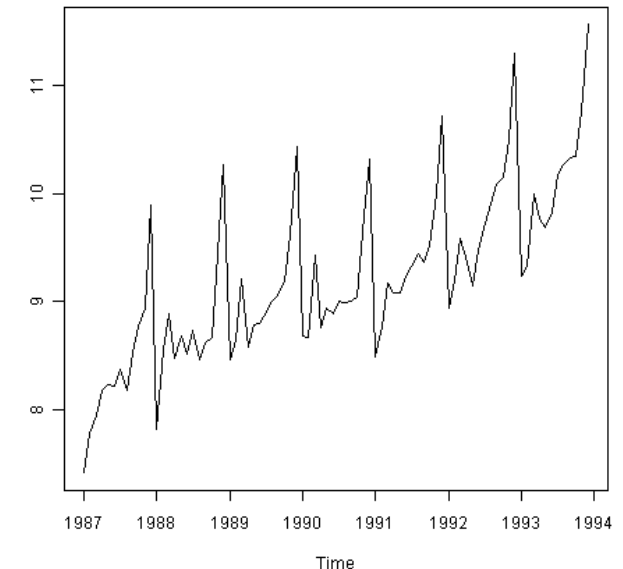


Other Data Mining Functions: Time and Ordering: Sequential Pattern, Trend and Evolution Analysis

- ❑ Sequence, trend and evolution analysis
 - ❑ Trend, time-series, and deviation analysis
 - ❑ e.g., regression and value prediction
 - ❑ Sequential pattern mining
 - ❑ e.g., buy digital camera, then buy large memory cards
 - ❑ Periodicity analysis
 - ❑ Motifs and biological sequence analysis
 - ❑ Approximate and consecutive motifs
 - ❑ Similarity-based analysis
- ❑ Mining data streams
 - ❑ Ordered, time-varying, potentially infinite, data streams

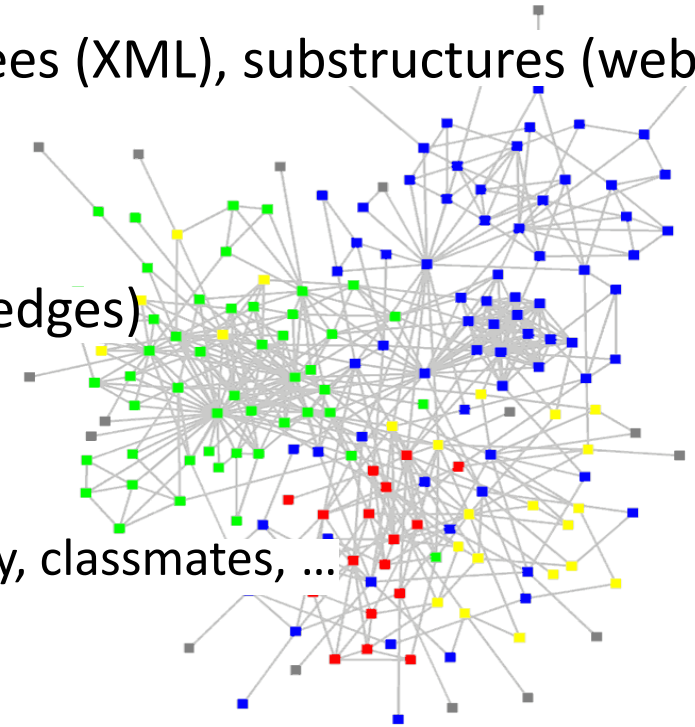


Image Source: <https://planopedia.com/wp-content/uploads/2014/10/trend-analysis.png>



Other Data Mining Functions: Structure and Network Analysis

- Graph mining
 - Finding frequent subgraphs (e.g., chemical compounds), trees (XML), substructures (web fragments)
- Information network analysis
 - Social networks: actors (objects, nodes) and relationships (edges)
 - e.g., author networks in CS, terrorist networks
 - Multiple heterogeneous networks
 - A person could be multiple information networks: friends, family, classmates, ...
 - Links carry a lot of semantic information: Link mining
- Web mining
 - Web is a big information network: from PageRank to Google
 - Analysis of Web information networks
 - Web community discovery, opinion mining, usage mining, ...



Evaluation of Knowledge

- ❑ Are all mined knowledge interesting?
 - ❑ One can mine tremendous amount of “patterns”
 - ❑ Some may fit only certain dimension space (time, location, ...)
 - ❑ Some may not be representative, may be transient, ...
- ❑ Evaluation of mined knowledge → directly mining only interesting knowledge?
 - ❑ Descriptive vs. predictive
 - ❑ Coverage
 - ❑ Typicality vs. novelty
 - ❑ Accuracy
 - ❑ Timeliness
 - ❑ ...

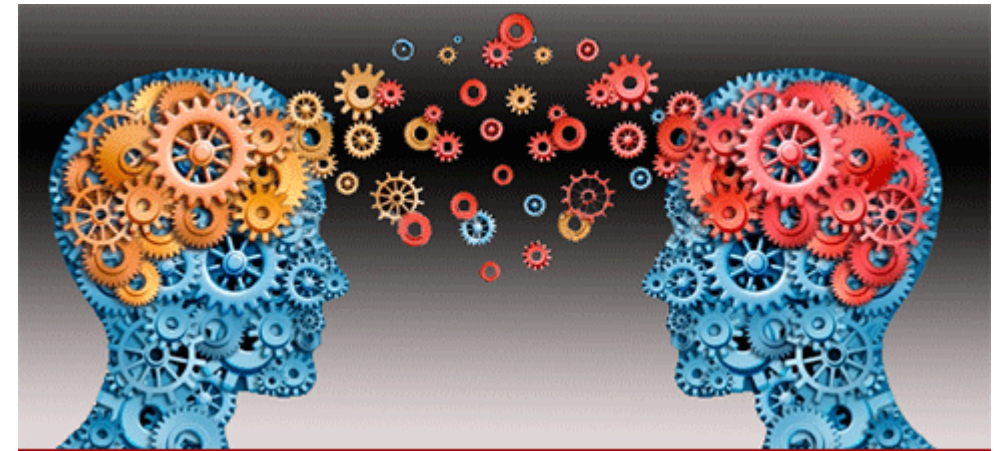


Image Source: http://ieg.worldbankgroup.org/Data/knowledge_banner.gif